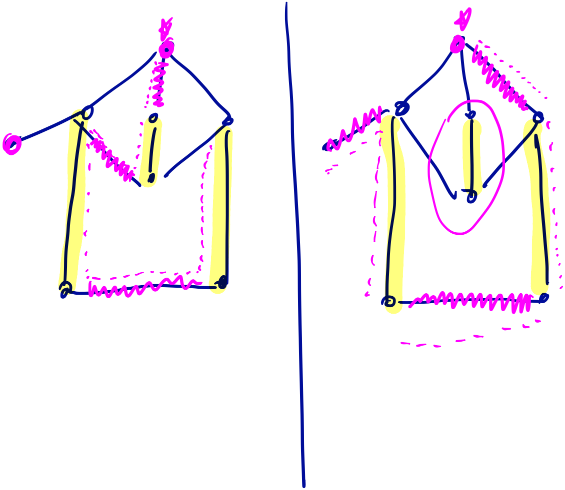
Finding an M-Augmenting Path



M-Alternating Trees

Given a matching M in $G=(V,E)$

let $X \subseteq V$ be nodes (vertices) not covered by M.

Also called "exposed".

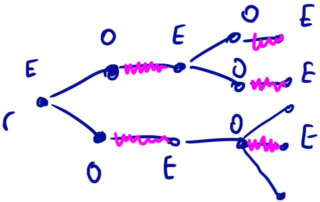
<u>Def'n</u>: M-alternating tree

A tree rooted in $r \in X$ s.t. along every path from root to a leaf, edges alternate not being in M and being in M.

<u>Observation</u>: If M-alt tree contains a vertex $v \in X \setminus \{r\}$ we have an M-aug path

Odd/Even and Maximal Trees

Def^n Given an M-alt tree, let Odd be the vertices that are an odd length from root.



Even be even-distance vertices : note: $r \in E$.

Def^n : A maximal M-alt tree is one where no even vertex has an edge in G to a vertex not in the tree; i.e., cannot add extra vertices.

Growing Maximal Trees

1. Choose root $r \in X$. Label it E.
2. Add all nodes adjacent to E vertices not visited so far. Call these odd.

If $\text{odd} \cap X \neq \emptyset$ STOP: found M-aug path.

3. For each $v \in \text{Odd}$ add matching edge (edge from M) and corresp vertex to E.

4. Once this terminates, remove all nodes & edges from G and repeat with next node in X.

A Sufficient Condition

Lemma: If above alg. terminates and

1. It did not find an M-aug path
2. There are no edges in G b/w vertices labeled "E",

Then M is a max size matching.